

Potential of GPT-4 for Detecting Errors in Radiology Reports: Implications for Reporting Accuracy

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See also the editorial by Forman in this issue.

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Background: Errors in radiology reports may occur because of resident-to-attending discrepancies, speech recognition inaccuracies, and large workload. Large language models, such as GPT-4 (ChatGPT; OpenAI), may assist in generating reports.

Purpose: To assess effectiveness of GPT-4 in identifying common errors in radiology reports, focusing on performance, time, and cost-efficiency.

Materials and Methods: In this retrospective study, 200 radiology reports (radiography and cross-sectional imaging [CT and MRI]) were compiled between June 2023 and December 2023 at one institution. There were 150 errors from five common error categories (omission, insertion, spelling, side confusion, and other) intentionally inserted into 100 of the reports and used as the reference standard. Six radiologists (two senior radiologists, two attending physicians, and two residents) and GPT-4 were tasked with detecting these errors. Overall error detection performance, error detection in the five error categories, and reading time were assessed using Wald χ^2 tests and paired-sample t tests.

Results: GPT-4 (detection rate, 82.7%; 124 of 150; 95% CI: 75.8, 87.9) matched the average detection performance of radiologists independent of their experience (senior radiologists, 89.3% [134 of 150; 95% CI: 83.4, 93.3]; attending physicians, 80.0% [120 of 150; 95% CI: 72.9, 85.6]; residents, 80.0% [120 of 150; 95% CI: 72.9, 85.6]; P value range, .522–.99). One senior radiologist outperformed GPT-4 (detection rate, 94.7%; 142 of 150; 95% CI: 89.8, 97.3; P = .006). GPT-4 required less processing time per radiology report than the fastest human reader in the study (mean reading time, 3.5 seconds $\pm$ 0.5 [SD] vs 25.1 seconds $\pm$ 20.1, respectively; P < .001; Cohen d = -1.08). The use of GPT-4 resulted in lower mean correction cost per report than the most cost-efficient radiologist (\$0.03 $\pm$ 0.01 vs \$0.42 $\pm$ 0.41; P < .001; Cohen d = -1.12).

Conclusion: The radiology report error detection rate of GPT-4 was comparable with that of radiologists, potentially reducing work hours and cost.

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For radiology reports to be effective, radiologists must ensure accuracy and consistency, particularly between the findings and impressions sections. Preliminary radiology reports are typically drafted by residents and subsequently reviewed and approved by board-certified radiologists. This legally necessary process increases accuracy but is time- and resource-intensive. In addition, increasing radiologist workloads (1), high-pressure clinical environments, and unreliable speech recognition (2,3) mean that radiology reports are prone to errors. A recent analysis of residents' preliminary reports (4) showed that the most common reporting errors are laterality discriminator confusions (left, right) and unnoticed descriptor misregistration by speech recognition (one, none). These two errors, if not corrected,

can have serious repercussions. Proofreading tools beyond basic spellcheck are not routinely available to check for these errors.

GPT-4 (ChatGPT; OpenAI) (5), an autoregressive large language model, may offer potential solutions to these challenges. Recent studies have demonstrated potential applications of GPT-4 in patient care (6–8). For instance, transforming free-text radiology reports into structured formats (9), automatically generating the impression section (10), and generating competent radiology reports (eg, for distal radius fractures) (11) underscore its utility in report standardization. The role of GPT-4 in education and knowledge expansion within the field of radiology has also been investigated. Its performance has

Summary

GPT-4 (ChatGPT; OpenAI) matched the performance of radiologists in detecting errors in radiology reports and has the potential to reduce processing time and cost.

Key Results

- Of 200 radiology reports (radiography and cross-sectional imaging), GPT-4 (ChatGPT; OpenAI) matched the average error detection performance of six radiologists (GPT-4, senior radiologists, attending physicians, and residents: 82.7% [124 of 150; 95% CI: 75.8, 87.9], 89.3% [134 of 150; 95% CI: 83.4, 93.3], 80.0% [120 of 150; 95% CI: 72.9, 85.6], and 80.0% [120 of 150; 95% CI: 72.9, 85.6], respectively; P value range, .522–.99).
- GPT-4 required less mean reading time per report than the fastest human reader in the study (3.5 seconds $\pm$ 0.5 [SD] vs 25.1 seconds $\pm$ 20.1, respectively; $P < .001$; Cohen $d = -1.08$).
- GPT-4 resulted in lower mean correction cost per report than the most cost-efficient radiologist (\$0.03 $\pm$ 0.01 vs \$0.42 $\pm$ 0.41, respectively; $P < .001$; Cohen $d = -1.12$).

been tested in radiology board examinations (12,13), providing insights into its strengths and limitations. Its capabilities in simplifying radiology reports show promising results (14–17). GPT-4 has also shown potential in extracting information from free-text CT reports (18), study and protocol selection (19–23), and in providing screening recommendations for breast cancer prevention (24). Its ability to generate differential diagnoses based on imaging patterns indicates its potential to aid in diagnostic processes (25,26).

Using GPT-4 to proofread radiology reports could alleviate the workload of supervising radiologists and be an educational resource for residents because it could correct radiology reports and point out errors. This would be particularly

beneficial if residents do not have access to tools that track the report modifications made by supervising radiologists (4). Therefore, this study aimed to evaluate the performance of GPT-4 in detecting the most common errors and discrepancies in radiology reports and to estimate its potential in reducing time and cost.

Materials and Methods

This retrospective study received ethical approval (23–1061-retro), and informed consent was waived because of its retrospective design. No patient-identifying information was provided to GPT-4.

Study Design and Data Set

A total of 200 original radiology reports from both radiography ($n = 89$) and cross-sectional imaging (CT and MRI; $n = 111$) covering a wide range of pathologic abnormalities from one tertiary care hospital (University Hospital Cologne, University of Cologne, Cologne, Germany) between June 2023 and December 2023 were selected and randomized by using a freeware research data randomization tool (<https://www.randomizer.org>). The data were randomized into two sets, correct and incorrect, that contained 100 reports each. Within the incorrect set, 150 errors were deliberately introduced by a radiology resident (R.H.), with a maximum of three errors per report (Fig 1).

The following five error categories were defined based on prior research to include the most common error types in radiology reports (2–4), as follows: (a) omission: the omission of relevant words or expressions, which encompasses both deletions and missing words (eg, fracture instead of no fracture); (b) insertion: the unintentional insertion of incorrect words or expressions, including inappropriate words, incorrect word substitutions, insertions, or word confusions (eg, abnormal instead of normal); (c) spelling errors: spelling mistakes, including word truncations, likely manual text processing by

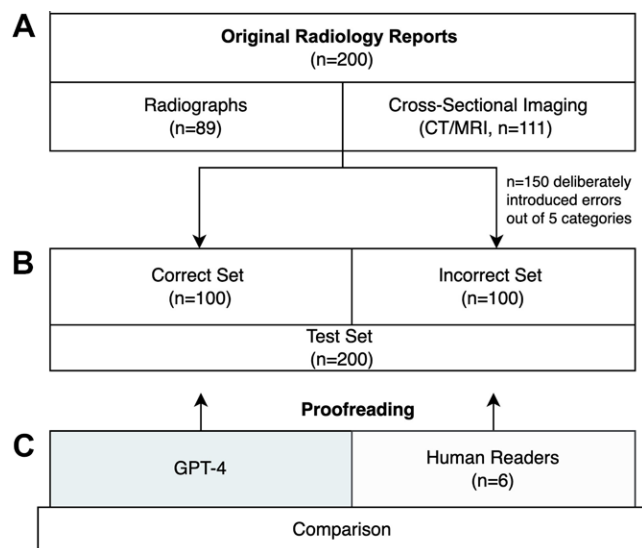


Figure 1: Study flowchart. (A) Initially, 200 original radiology reports from radiographs and cross-sectional imaging (CT and MRI) studies were selected. (B) These were then randomized into two sets: a correct set and an incorrect set, each containing 100 radiology reports. Within the incorrect set, 150 errors across five categories (omission, insertion, spelling errors, side confusion, other errors) were deliberately introduced, with a maximum of three errors per case. (C) GPT-4 and six radiologists were tasked with evaluating each radiology report to identify potential errors, allowing for a comparative analysis of their performance.

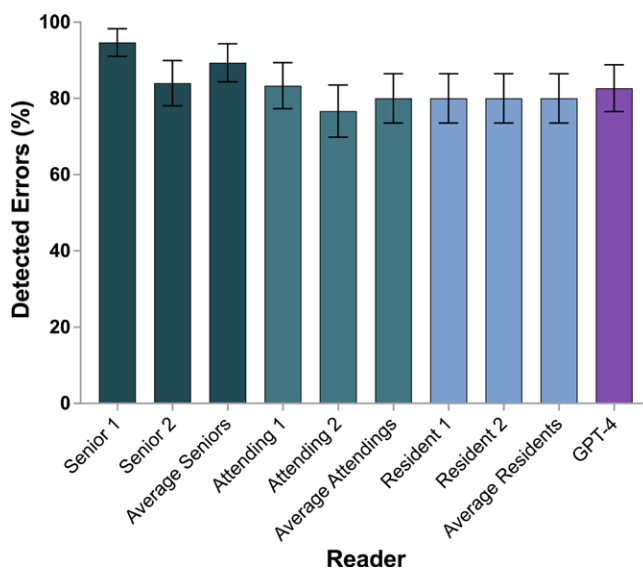


Figure 2: Bar graph shows the percentage of detected errors for GPT-4 and the radiologists. The error bars are 95% CIs.

#	Radiological Report	Error and Error Type(s)	Error Detection	
			Error 1	Error 2
1	Findings: A conventional radiograph from June 14, 2021, is used for comparison. The cardiac silhouette is centrally aligned and unchanged. The mediastinum is unremarkable. Signs of aortic sclerosis. The right diaphragm is elevated, due to known resection of the lower lobe. Regressing pleural effusion on the right side, with associated atelectasis of the middle lobe. No pleural effusion on the left side. No evidence of pneumothorax or pneumonic infiltrates. The diaphragm is well-positioned with a sharp costophrenic angle. No abnormalities are noted in the soft tissues or in the visualized upper abdomen. Impression: Progressing pleural effusion on the right side, without associated atelectasis.	Error: The 'Findings' describe a resolving right-sided pleural effusion with accompanying atelectasis of the middle lobe. However, the 'Impression' notes a worsening right-sided pleural effusion without atelectasis of the middle lobe. Error type(s): Spelling error (Regressing vs progressing) Omission (with vs without)	GPT-4: yes Seniors: R1: yes R2: yes Attending: R3: yes R4: yes Residents: R5: yes R6: yes	GPT-4: yes Seniors: R1: yes R2: yes Attending: R3: yes R4: yes Residents: R5: yes R6: yes
2	Findings: Field strength: 3.0 Tesla. No relevant prior images available. Prostate evaluation: Adenomatous hyperplasia of the transition zone with multiple adenomatous nodules. The prostate measures 5.5 cm × 4.5 cm in the axial plane and 4.4 cm in the craniocaudal extent, resulting in a volume of 56.63 mL. The transition zone measures 4.8 cm × 4.0 cm in the axial plane and 4.0 cm in the craniocaudal extent, resulting in a volume of 39.94 mL. Index lesion(s): Left anterior transition zone (TZa), basal (Series/Image number: 501/17); focal, maximum diameter: approximately 13 mm. T2 signal: hypointense. DWI: hyperintense. ADC: hypointense. DCE: hyperperfused. PI-RADS: 3. Lymph Nodes: No suspicious lymph nodes in the examined area. Bladder: unremarkable. Rectum: unremarkable. Bones: No suspicious bone lesions. Impression: The mpMRI reveals a PI-RADS 3 index lesion in the posterior basal transition zone on the left.	Error: The 'Findings' describe an index lesion in the basal left anterior transition zone. However, the 'Impression' states an index lesion in the posterior basal transition zone on the left. Error type(s): Side confusion (anterior vs posterior).	GPT-4: no Seniors: R1: yes R2: yes Attending: R3: yes R4: yes Residents: R5: yes R6: yes	
3	Findings: No prior images available for comparison. There is a fracture at the inferior pole of the patella with displacement of the larger fragment inferiorly and anteriorly, and displacement of a smaller bony fragment inferiorly. Alignment within the right knee joint is appropriate. No further pathologic step-offs or interruptions of the cortex are delineated. The bone trabeculae are homogeneously structured. The visualized soft tissues are unremarkable. Impression: Nondisplaced fracture at the superior pole of the patella.	Error: The 'Findings' describe a displaced fracture at the inferior pole of the patella. However, the 'Impression' states a nondisplaced fracture at the superior pole of the patella. Error type(s): Insertion (displaced vs nondisplaced). Side confusion (inferior vs superior).	GPT-4: yes Seniors: R1: yes R2: yes Attending: R3: yes R4: yes Residents: R5: yes R6: yes	GPT-4: yes Seniors: R1: yes R2: yes Attending: R3: yes R4: no Residents: R5: yes R6: yes

Figure 3: Comparative proofreading examples by GPT-4 and the human readers show incorrect radiology reports with the respective errors and error types and the corresponding proofreading results. ADC = apparent diffusion coefficient, DCE = dynamic contrast enhanced, DWI = diffusion-weighted imaging, mpMRI = multiparametric MRI, PI-RADS = Prostate Imaging Reporting and Data System, R1 = reader 1, R2 = reader 2, R3 = reader 3, R4 = reader 4.

radiologists through typing errors or inaccurate selection of text that is to be removed or edited (eg, pneumothorax instead of pneumothorax); (d) side confusion (eg, right instead of left, lateral instead of medial); and (e) other errors: errors that did not fit into the above categories, including incorrect date entries, errors in numbering of images and/or series, mistakes in units of measurement (eg, centimeter vs millimeter), template errors, and punctuation mistakes.

Only the errors intentionally inserted into the text were used as a reference standard. To ensure that no other errors were present in the text of the radiology reports, they were verified by three readers (J.K., R.J.G., and R.H., with 5, 5, and 3 years of

experience in training, respectively). Discrepancies were resolved in a consensus reading with all three readers.

Error and Time Calculations

Human readers.—Four board-certified radiologists with different levels of clinical experience, composed of two experienced senior radiologists (A.C.B. and T.P., with 17 and 20 years of experience, respectively) two attending radiologists (A.I.I. and L.P., with 8 and 7 years of experience, respectively), and two residents in training (C.H.G. and P.F., with 5 years of experience, respectively), evaluated each radiology report for

Table 1: Comparison of Error Detection Rates between GPT-4 and the Radiologists

Reader	Total		Radiography		CT and MRI	
	Detection Rate (%)	<i>P</i> Value*	Detection Rate (%)	<i>P</i> Value	Detection Rate (%)	<i>P</i> Value
Senior 1	94.7 (89.8, 97.3) [142/150]	.006	90 (80, 95) [54/60]	>.99	98 (92, 99) [88/90]	.002
Senior 2	84.0 (77.3, 89.0) [126/150]	>.99	82 (70, 89) [49/60]	>.99	86 (77, 91) [77/90]	>.99
Senior radiologist average	89.3 (83.4, 93.3) [134/150]	.522	86 (75, 93) [51.5/60]	>.99	92 (84, 96) [82.5/90]	.200
Attending 1	83.3 (76.6, 88.4) [125/150]	>.99	83 (72, 91) [50/60]	>.99	83 (74, 90) [75/90]	>.99
Attending 2	76.7 (69.3, 82.7) [115/150]	>.99	73 (61, 83) [44/60]	.404	79 (69, 86) [71/90]	>.99
Attending physician average	80.0 (72.9, 85.6) [120/150]	>.99	78 (66, 87) [47/60]	>.99	81 (72, 88) [73/90]	>.99
Resident 1	80.0 (72.9, 85.6) [120/150]	>.99	78 (66, 87) [47/60]	>.99	81 (72, 88) [73/90]	>.99
Resident 2	80.0 (72.9, 85.6) [120/150]	>.99	75 (63, 84) [45/60]	.453	83 (74, 90) [75/90]	>.99
Resident average	80.0 (72.9, 85.6) [120/150]	>.99	77 (65, 86) [46/60]	.543	82 (73, 89) [74/90]	>.99
GPT-4	82.7 (75.8, 87.9) [124/150]		85 (74, 92) [51/60]		81 (72, 88) [73/90]	

Note.—Data in parentheses are 95% CIs; data in brackets are numerators/denominators. Bonferroni correction was used to correct *P* values for multiple comparisons.

* The number of correctly detected errors by GPT-4 was compared with the radiologists by using Wald χ^2 tests.

potential errors. Using a digital stopwatch, they measured the length of time to evaluate each report.

GPT-4.—GPT-4 (<https://openai.com/research/gpt-4>) (5) was accessed via the application programming interface (OpenAI; <https://platform.openai.com/docs/api-reference/introduction>). Reports were collected (Excel version 16.8; Microsoft) and individually evaluated using the application programming interface via a Python script (version 3.11; Python Software Foundation).

Similar to the human readers, GPT-4 was prompted with evaluating each radiology report for potential errors using zero-shot prompting (27). This method involves giving GPT-4 a single instruction without subsequent modifications, unlike single-shot or chain-of-thought prompting, where GPT-4's responses are refined through an iterative, example-inclusive approach. The prompt was as follows: "In the following, I will provide you with a radiology report, divided into a 'findings' and an 'impression' section. Please evaluate the report for mistakes and assess and validate the consistency between the 'findings' and the 'impression' sections, highlighting any discrepancies or notable points."

The time required for each radiology report correction by GPT-4 was evaluated by measuring the duration from when the prompt was sent to when the final response was received. This measurement was performed in 15 randomly chosen reports of varying length.

Cost Calculation

Human readers.—In the cost analysis, expressed in U.S. dollars per time unit, we referenced the collective labor agreements that are nationally recognized in Germany (<https://www.marburger-bund.de/bundesverband/tarifvertraege>). These agreements indicate the salary for predetermined monthly working hours. For the conversion of cost from euros to U.S. dollars, the current exchange rate was used (as of January 2, 2024).

GPT-4.—When this study was conducted, GPT-4 cost was set at \$0.06 for every 1000 tokens used in a prompt and \$0.12 for every 1000 tokens overall. The specific cost may vary depending on the nature of the task. Based on the cost analysis of 15 randomly chosen radiology reports of varying length, the total cost of the evaluation performed in this study was calculated.

Statistical Analysis

All analyses were performed using software (IBM SPSS Statistics version 26.0, IBM; and Python version 3.11, Python). We evaluated the number of correctly detected errors and the time to process each radiology report as outcome measures. The number of errors detected correctly by GPT-4 was compared with that of the radiologists using Wald χ^2 tests. The Wilson method was used to calculate 95% CIs (28). This analysis was performed for all modalities combined and separately for radiographs and cross-sectional imaging. The average time and cost to process a GPT-4-corrected radiology report was compared with that of the radiologists using paired-sample *t* tests.

Bonferroni correction was used to correct for multiple comparisons. Two-sided *P* < .05 indicated a statistically significant difference. Cohen *d* was used as the effect size for the differences in processing time, with *d* values of 0.2, 0.5, and 0.8 constituting small, medium, and large effects, respectively (29). Interrater agreement was assessed using Cohen κ , as follows: 0.01–0.20, none to slight; 0.21–0.40, fair; 0.41–0.60, moderate; 0.61–0.80, substantial; and 0.81–1.00, almost perfect agreement (30). Because of the nature of the research and the lack of prior effect sizes for comparison, no power analysis was performed.

Results

Performance in Detecting Errors

In the overall analysis, GPT-4 detected fewer errors compared with the best-performing senior radiologist (detection rate, 82.7% [124 of 150; 95% CI: 75.8, 87.9] vs 94.7% [142 of

Table 2: Comparison of Detection Rates for Different Error Types in Radiology Reports

Reader	Omission		Insertion		Spelling		Side Confusion		Other	
	Detection Rate (%)	<i>P</i> Value*	Detection Rate (%)	<i>P</i> Value	Detection Rate (%)	<i>P</i> Value	Detection Rate (%)	<i>P</i> Value	Detection Rate (%)	<i>P</i> Value
Senior 1	90 (75, 97) [28/31]	>.99	91 (78, 96) [39/43]	.653	100 (83, 100) [19/19]	.532	97 (84, 99) [31/32]	.199	100 (87, 100) [25/25]	>.99
Senior 2	81 (64, 91) [25/31]	>.99	79 (65, 89) [34/43]	>.99	95 (75, 99) [18/19]	>.99	84 (68, 93) [27/32]	>.99	88 (70, 96) [22/25]	>.99
Senior radiologist average	86 (69, 94) [26.5/31]	>.99	85 (71, 93) [36.5/43]	>.99	97 (79, 100) [18.5/19]	>.99	91 (76, 97) [29/32]	>.99	94 (78, 99) [23.5/25]	>.99
Attending 1	74 (57, 86) [23/31]	>.99	86 (73, 93) [37/43]	>.99	84 (62, 95) [16/19]	>.99	91 (76, 97) [29/32]	.792	80 (61, 91) [20/25]	.756
Attending 2	65 (47, 79) [20/31]	.392	77 (62, 87) [33/43]	>.99	90 (69, 97) [17/19]	>.99	75 (58, 87) [24/32]	>.99	84 (65, 94) [21/25]	>.99
Attending physician average	69 (52, 83) [21.5/31]	.615	81 (67, 90) [35/43]	>.99	87 (66, 96) [16.5/19]	>.99	83 (67, 92) [26.5/32]	>.99	82 (63, 92) [20.5/25]	.809
Resident 1	77 (60, 89) [24/31]	>.99	79 (65, 89) [34/43]	>.99	84 (62, 95) [16/19]	>.99	78 (61, 89) [25/32]	>.99	84 (65, 94) [21/25]	>.99
Resident 2	55 (38, 71) [17/31]	.003	81 (67, 90) [35/43]	>.99	90 (69, 97) [17/19]	>.99	100 (89, 100) [32/32]	.025	76 (57, 89) [19/25]	.371
Resident average	66 (49, 80) [20.5/31]	.253	80 (66, 89) [34.5/43]	>.99	87 (66, 96) [16.5/19]	>.99	89 (74, 96) [28.5/32]	>.99	80 (61, 91) [20/25]	.262
GPT-4	84 (67, 93) [26/31]		77 (62, 87) [33/43]		84 (62, 95) [16/19]		78 (61, 89) [25/32]		96 (81, 99) [24/25]	

Note.—Data in parentheses are 95% CIs; data in brackets are numerators/denominators. Other errors included errors that did not fit into the defined categories, such as incorrect date entries, errors in numbering of images and/or series, and mistakes in specifying units of measurement (eg, centimeter vs millimeter). Bonferroni correction was used to correct *P* values for multiple comparisons.

* The number of errors correctly detected by GPT-4 was compared with the radiologists by using Wald χ^2 tests.

150; 95% CI: 89.8, 97.3]; $P = .006$). However, there was no evidence of a difference in the percentage of average performance in error detection rate between GPT-4 and all the other radiologists per report (GPT-4, senior radiologists, attending physicians, and residents: 82.7% [124 of 150; 95% CI: 75.8, 87.9], 89.3% [134 of 150; 95% CI: 83.4, 93.3], 80.0% [120 of 150; 95% CI: 72.9, 85.6], and 80.0% [120 of 150; 95% CI: 72.9, 85.6], respectively; P value range, .522–.99) (Figs 2, 3; Table 1).

In the subgroup analysis of different imaging modalities, there was no evidence of a difference in detection of errors in radiography reports when comparing GPT-4 with the best-performing senior radiologist (detection rate, 85% [51 of 60; 95% CI: 74, 92] vs 90% [54 of 60; 95% CI: 80, 95], respectively; $P > .99$) or the performance of other radiologists (P value range, .40–.99) (Table 1). Regarding errors in CT and MRI reports, GPT-4 detected fewer errors than the best-performing senior radiologist (detected errors, 81% [73 of 90; 95% CI: 72, 88] vs 98% [88 of 90; 95% CI: 92, 99]; $P = .002$). However, there was no evidence of a difference in the percentage of detected errors between GPT-4 and the other radiologists per report (P value range, .200–.99) (Table 1).

The performance of GPT-4 in detecting side confusion was worse than the best-performing radiologist (detection rate,

78% [25 of 32; 95% CI: 61, 89] vs 100% [32 of 32; 95% CI: 89, 100]; $P = .025$) (Table 2). However, there was no evidence of a difference in the percentage of detected errors between GPT-4 and all the other radiologists per report (P value range, .199–.99). Regarding the other error categories, there was no evidence of GPT-4 detecting fewer errors compared with the radiologists (P value range, .532–.99).

Of the 200 reports, GPT-4 incorrectly labeled eight reports as containing errors that did not contain any errors. However, there was no evidence of a difference between GPT-4 and the radiologists when comparing incorrectly flagged radiology reports (P value range, .064–.99) (Table 3).

Interrater Agreement

Interrater agreement between GPT-4 and the radiologists and among the radiologists was only slight to fair (Cohen $\kappa = 0.02$ – 0.34), indicating that there was no specific pattern in the detection of errors (Table 4).

Reading Time

The total reading time for all 200 reports for GPT-4 was 0.19 hours; the fastest radiologist read the 200 reports in 1.4 hours, and the slowest radiologist read the reports in 5.74 hours (Fig 4A). The mean reading time per radiology report for GPT-4

was faster than the fastest radiologist (mean reading time, 3.5 seconds $\pm$ 0.5 [SD] vs 25.1 seconds $\pm$ 20.1; $P < .001$; Cohen $d = -1.08$) (Fig 4B, Table 5).

Cost

The total estimated average cost for proofreading all 200 reports by the six human readers was \$190.17 (average cost per senior radiologist, \$231.85; average cost per attending physician, \$156.89; and average cost per resident, \$181.98) compared with \$5.78 for GPT-4 (Fig 4C).

The total estimated average cost for proofreading one radiology report by the six human readers was \$0.96 (average cost per senior radiologist, \$1.17; average cost per attending physician, \$0.78; average cost per resident, \$0.93) compared with \$0.03 for GPT-4. The mean cost per report for GPT-4 was lower than that of the radiologist who had the lowest mean cost per report (mean cost per report, \$0.03 $\pm$ 0.01 vs \$0.42 $\pm$ 0.41, respectively; $P < .001$; Cohen $d = -1.12$) (Fig 4D).

Table 3: Comparison of Incorrectly Flagged Radiology Reports between GPT-4 and the Radiologists

Reader	Rate of Incorrectly Flagged Reports (% Rate)	<i>P</i> Value*
Senior 1	0.5 (0.1, 2.8) [1/200]	.064
Senior 2	0.5 (0.1, 2.8) [1/200]	.162
Senior radiologist average	0.5 (0.1, 2.8) [1/200]	.084
Attending 1	0.5 (0.1, 2.8) [1/200]	.064
Attending 2	1.5 (0.5, 4.3) [3/200]	>.99
Attending physician average	1.0 (0.3, 3.6) [2/200]	.287
Resident 1	1.0 (0.3, 3.6) [2/200]	.500
Resident 2	1.0 (0.3, 3.6) [2/200]	.287
Resident average	1.0 (0.3, 3.6) [2/200]	.287
GPT-4	4.0 (2.0, 7.7) [8/200]	

Note.—Data in parentheses are 95% CIs; data in brackets are numerators/denominators. Bonferroni correction was used to correct *P* values for multiple comparisons.

* The number of incorrectly flagged reports by GPT-4 was compared with the radiologists by using Wald χ^2 tests.

Discussion

Creating accurate and consistent radiology reports is crucial for patient care and treatment decisions. This process is challenged by resident-to-attending discrepancies, inaccuracies of speech recognition software, large workload, and time constraints. These challenges may lead to errors and discrepancies in radiology reports. The advanced text-processing capabilities of large language models, such as GPT-4, have the potential to enhance the report generation process. Therefore, we aimed to evaluate the performance of GPT-4 in detecting the most common errors and discrepancies in radiology reports and estimate its potential to reduce time and cost. For that purpose, the performance of GPT-4 in detecting 150 predefined errors and discrepancies introduced into 200 radiology reports was measured and compared with that of six radiologists of different experience levels. Our results demonstrate that the proofreading performance of GPT-4 is comparable with human readers, independent of the human readers' clinical experience (GPT-4 vs average senior radiologists detection rate, 82.7% vs 89.3% [$P > .99$]; GPT-4 vs average attending physicians, 82.7% vs 80.0% [$P > .99$]; GPT-4 vs average residents, 82.7% vs 80.0%; $P > .99$). The performance of GPT-4 was only worse than that of one senior radiologist with 17 years of experience (detection rate, 82.7% vs 94.7%; $P = .006$). GPT-4 demonstrated a shorter processing time than any human reader, and the mean reading time per report for GPT-4 was faster than the fastest radiologist in the study (mean reading time, 3.5 seconds $\pm$ 0.5 vs 25.1 seconds $\pm$ 20.1; $P < .001$; Cohen $d = -1.08$). The total estimated average cost for proofreading 200 reports by the six human readers was \$190.17 (average senior radiologists, \$231.85; average attending physicians, \$156.89; and average residents, \$181.98) compared with \$5.78 for GPT-4.

Our results suggest that large language models may perform the task of radiology report proofreading with a level of proficiency comparable to that of most human readers. Because errors in radiology reports are likely to occur across all experience levels (2,4), our findings may be reflective of typical clinical environments. This underscores the potential of artificial intelligence to improve radiology workflows beyond image interpretation (31,32). Regarding time savings, our results align with other studies investigating the implementation of large language models into radiologic workflows (23,33). When used

Table 4: Interrater Agreement between GPT-4 and the Radiologists

Reader	Senior 1	Senior 2	Attending 1	Attending 2	Resident 1	Resident 2	GPT-4
Senior 1	...	0.12	0.11	0.11	0.02	0.14	0.04
Senior 2	0.12	...	0.15	0.18	0.14	0.10	0.04
Attending 1	0.11	0.15	...	0.34	0.13	0.31	0.17
Attending 2	0.11	0.18	0.34	...	0.24	0.24	0.24
Resident 1	0.02	0.14	0.13	0.24	...	0.21	0.25
Resident 2	0.14	0.10	0.31	0.24	0.21	...	0.25
GPT-4	0.04	0.04	0.17	0.24	0.25	0.25	...

Note.—Data are Cohen κ values (0.01–0.20, none to slight agreement; 0.21–0.40, fair agreement; 0.41–0.60, moderate agreement; 0.61–0.80, substantial agreement; and 0.81–1.00, almost perfect agreement (30)).

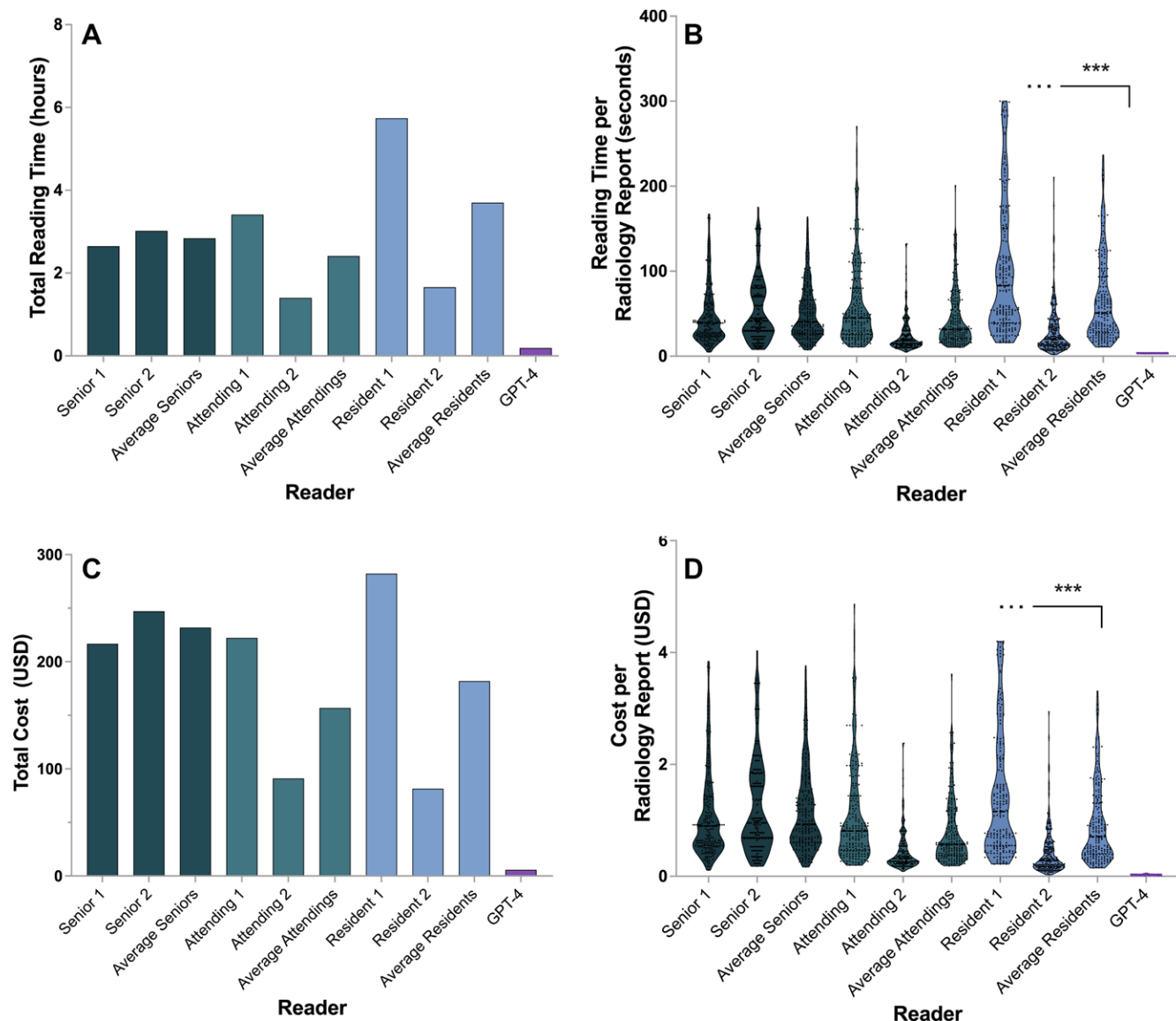


Figure 4: (A) Bar graph shows total reading time in seconds, (B) Violin plot shows reading time per radiology report in seconds, (C) bar graph shows total cost in U.S. dollars, and (D) violin plot shows cost per radiology report in U.S. dollars. Dashed lines are the medians and dotted lines are quartiles. *** $P < .001$.

as a proofreading tool, GPT-4 is potentially more cost-effective compared with human readers, with similar performance levels. This effect is primarily because of the time expenditure and the individual salary of the radiologists. However, previous research (34–36) in radiology indicates that human performance might decrease due to the demands of multitasking and off-hours shifts, whereas GPT-4 might offer consistent performance.

Our study had limitations. First, this was an experimental study setting including a correct and an incorrect set of radiology reports, with 150 deliberately introduced errors in the latter. Although the introduced errors and error categories are based on common error patterns identified in existing literature, it must be noted that the methodical categorization of errors fails to capture the wide spectrum of errors that are present in genuine radiology reports (4). Consequently, the actual type and likelihood of potential errors occurring, which might also depend on the experience level of the radiologist who created

the report, are not adequately represented in our data set. Second, regarding the time reduction due to the implementation of GPT-4, we refrained from a direct comparison between uncorrected and GPT-4-corrected radiology reports because of the risk of biased results from insufficient blinding of the readers. Third, the conclusion that financial resources can be saved by using GPT-4 has several shortcomings. The cost observed in our experimental setup falls short of capturing the true cost of implementing a large language model into the clinical workflow by neglecting the additional cost of infrastructure and data protection. Furthermore, the error detection rate of GPT-4 requires human validation: this is not only necessary but also legally mandated. Fourth, the experimental study setting might induce a Hawthorne effect (37) where participants alter their behavior because of their awareness of being observed, potentially leading to a higher detection rate than what might occur in a typical clinical setting. Furthermore, because of the scope

Table 5: Mean Reading Time and Cost Per Radiology Report

Reader	Mean Reading Time			Mean Cost		
	Per Radiology Report (sec)	Cohen <i>d</i>	<i>P</i> Value*	Per Radiology Report (\$)	Cohen <i>d</i>	<i>P</i> Value
Senior 1	47.7 ± 34.1	-1.30	<.001	1.10 ± 0.78	-1.36	<.001
Senior 2	54.4 ± 38.0	-1.34	<.001	1.25 ± 0.87	-1.40	<.001
Senior radiologist average	51.1 ± 32.3	-1.47	<.001	1.17 ± 0.74	-1.54	<.001
Attending 1	61.3 ± 49.0	-1.36	<.001	1.10 ± 0.88	-1.22	<.001
Attending 2	25.1 ± 20.1	-1.08	<.001	0.45 ± 0.36	-1.17	<.001
Attending physician average	43.2 ± 32.3	-1.23	<.001	0.78 ± 0.58	-1.29	<.001
Resident 1	103.3 ± 79.4	-1.26	<.001	1.44 ± 1.11	-1.28	<.001
Resident 2	29.8 ± 29.2	-0.90	<.001	0.42 ± 0.41	-1.12	<.001
Resident average	66.6 ± 50.5	-1.25	<.001	0.93 ± 0.71	-1.28	<.001
GPT-4	3.5 ± 0.5	...	...	0.03 ± 0.01	...	...

Note.—Mean data are ± SDs. Cohen *d* was used as the effect size for the differences in reading time and cost with *d* values of 0.2, 0.5, and 0.8 constituting small, medium, and large effects, respectively. The average time and cost to process a radiology report by GPT-4 was compared with the radiologists using paired-sample *t* tests. Bonferroni correction was used to correct *P* values for multiple comparisons.

of our study design, the potential effects of domain-specific fine-tuning of the performance of GPT-4 and its applicability into existing workflows remain unclear. These limitations show that it is necessary for further studies to investigate how the immediate error feedback of GPT-4 in a clinical setting might enhance the overall quality of findings in a radiology report.

In conclusion, GPT-4 demonstrated an error detection rate in radiology reports that was comparable to radiologists, potentially saving work hours and reducing costs. Nevertheless, legal and privacy considerations and a remaining performance gap compared to theoretically flawless radiology reports emphasize the ongoing need for human supervision in the report generation process. To implement a large language model into the local hospital information system, data privacy must be ensured. An offline or local private server-based version of a large language model, potentially equipped with a corresponding application programming interface, is a promising solution. Such a system may be an educational asset, serving as an interactive tutor for residents, highlighting common errors, and providing insights fostering a robust learning environment, allowing for real-time analysis and correction of radiology reports and immediate feedback regarding errors (38).

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